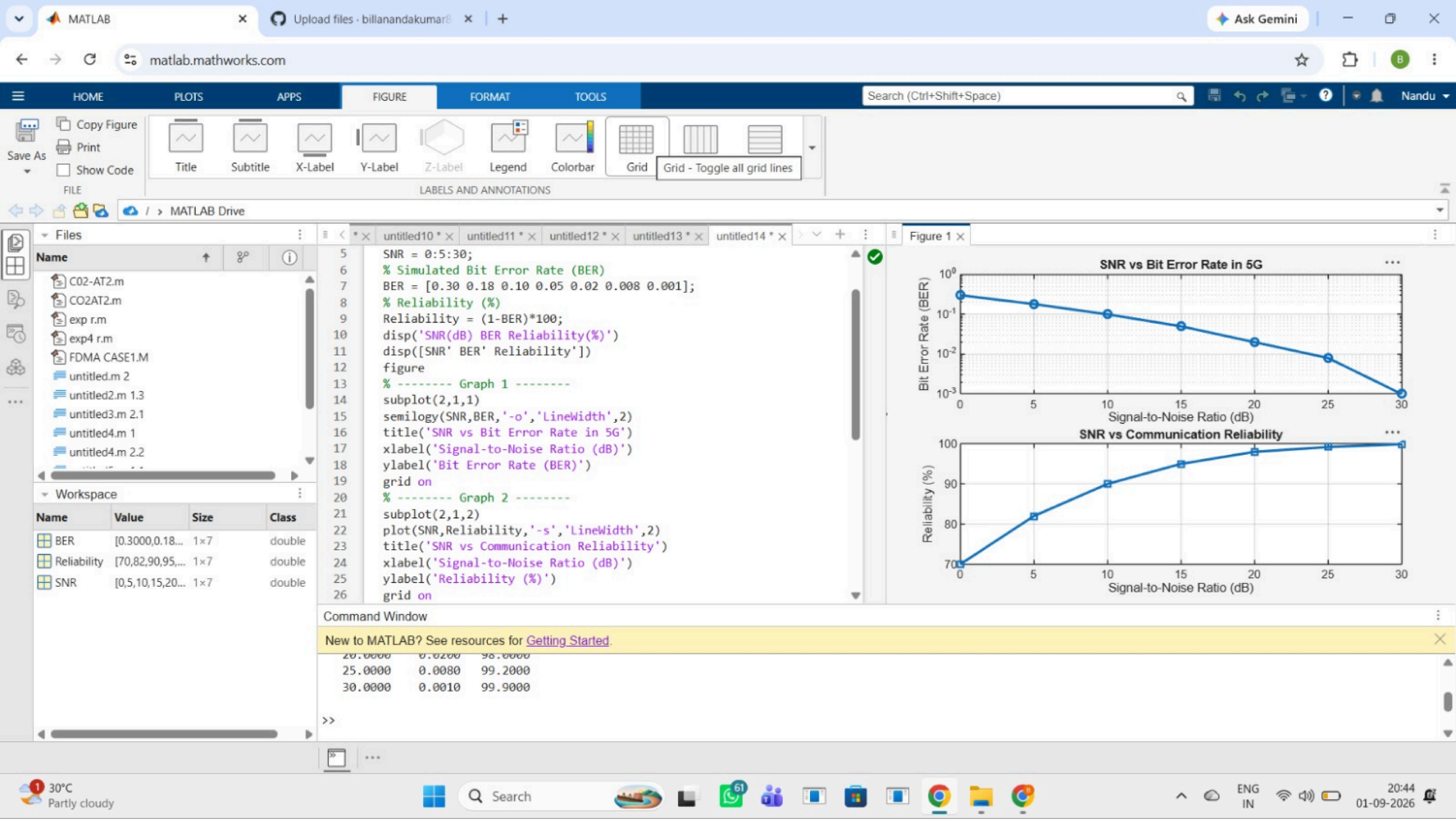
MATLAB
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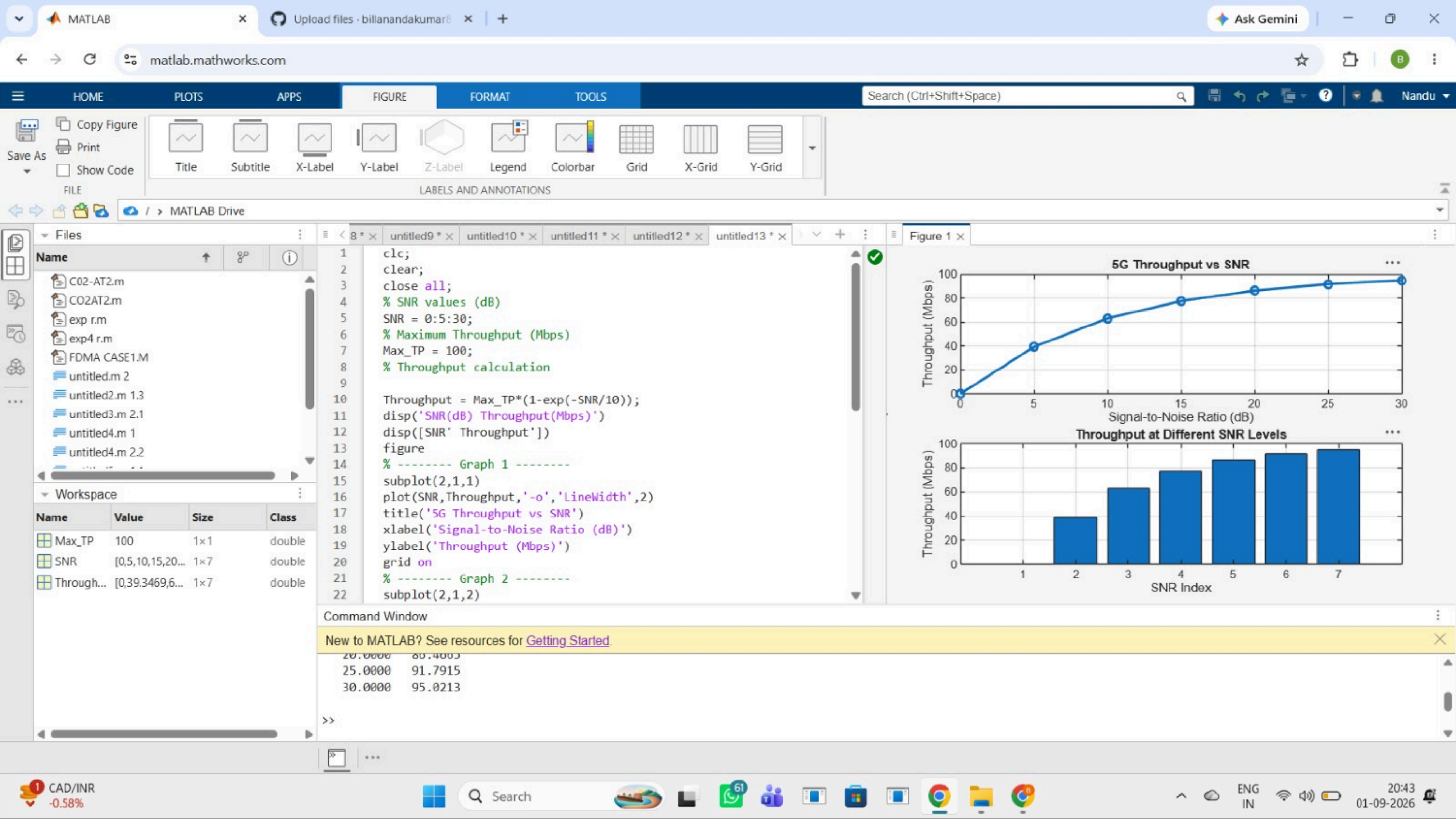
Grid - Toggle all grid lines
MATLAB Drive
Files
C02-AT2.m
CO2AT2.m
exp r.m
exp4 r.m
FDMA CASE1.M
Workspace
BER [0.3000,0.18... 1x7 double
Reliability [70,82,90,95,... 1x7 double
SNR [0,5,10,15,20... 1x7 double
SNR = 0:5:30;
% Simulated Bit Error Rate (BER)
BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];
% Reliability (%)
Reliability = (1-BER)*100;
disp('SNR(dB) BER Reliability(%)')
disp([SNR' BER' Reliability'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
semilogy(SNR,BER,'-o','LineWidth',2)
title('SNR vs Bit Error Rate in 5G')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Bit Error Rate (BER)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
plot(SNR,Reliability,'-s','LineWidth',2)
title('SNR vs Communication Reliability')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Reliability (%)')
grid on
Figure 1
SNR vs Bit Error Rate in 5G
Bit Error Rate (BER)
Signal-to-Noise Ratio (dB)
SNR vs Communication Reliability
Reliability (%)
Signal-to-Noise Ratio (dB)
Command Window
New to MATLAB? See resources for Getting Started.
25.0000 0.0080 99.2000
30.0000 0.0010 99.9000
30°C Partly cloudy
20:44
01-09-2026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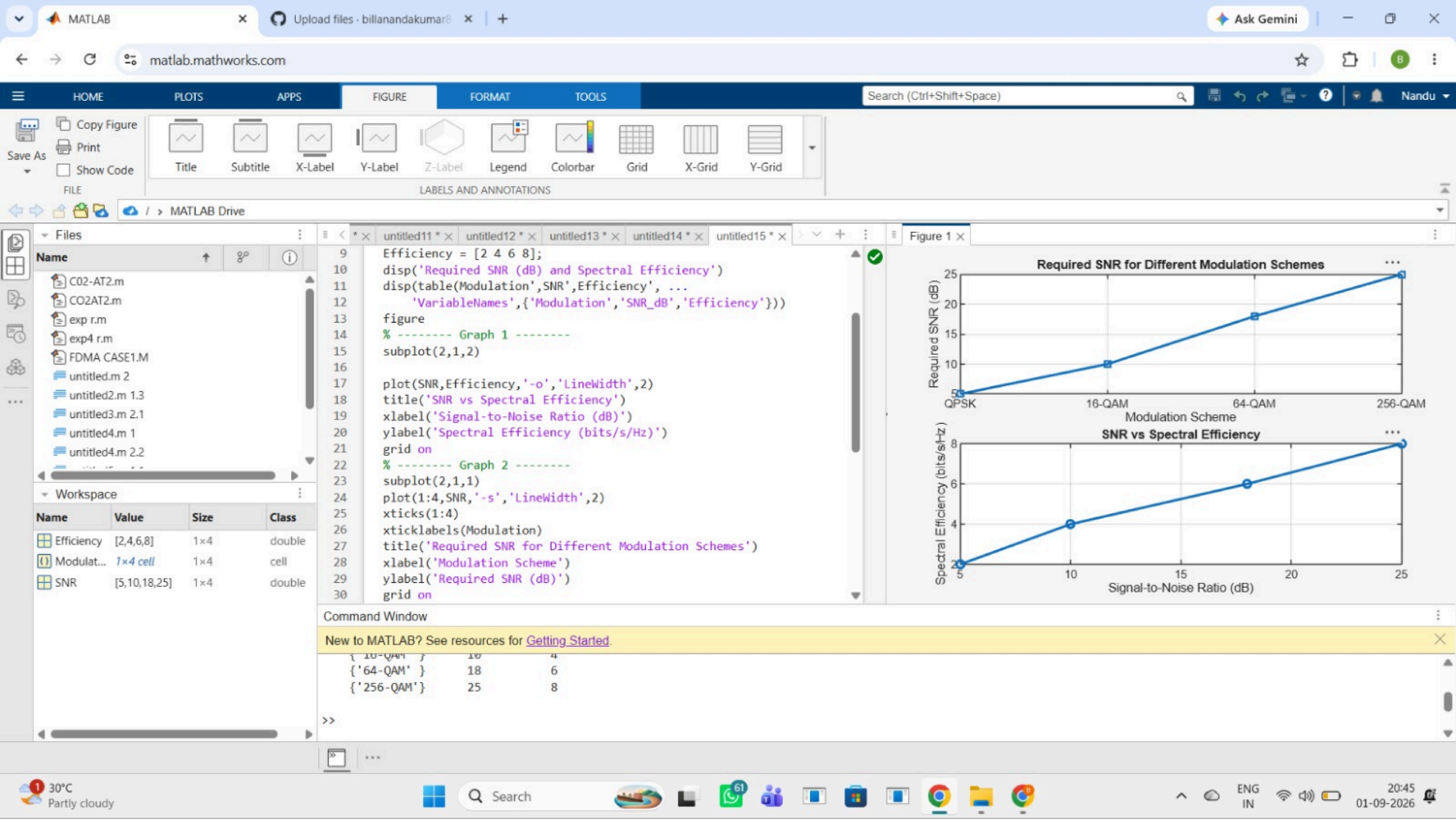
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
MATLAB Drive
Files
C02-AT2.m
CO2AT2.m
exp r.m
exp4 r.m
FDMA CASE1.M
untitled.m 2
untitled2.m 1.3
untitled3.m 2.1
untitled4.m 1
untitled4.m 2.2
Workspace
Name Value Size Class
Max_TP 100 1×1 double
SNR [0,5,10,15,20... 1×7 double
Through... [0,39.3469,6... 1×7 double
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Maximum Throughput (Mbps)
Max_TP = 100;
% Throughput calculation
Throughput = Max_TP*(1-exp(-SNR/10));
disp('SNR(dB) Throughput(Mbps)')
disp([SNR' Throughput'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR,Throughput,'-o','LineWidth',2)
title('5G Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
Figure 1
5G Throughput vs SNR
Throughput (Mbps)
Signal-to-Noise Ratio (dB)
Throughput at Different SNR Levels
Throughput (Mbps)
SNR Index
Command Window
New to MATLAB? See resources for Getting Started.
25.0000 91.7915
30.0000 95.0213
>>
CAD/INR -0.58%
Search
ENG IN
20:43
01-09-2026



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exp4 r.m
FDMA CASE1.M
untitled.m 2
untitled2.m 1.3
untitled3.m 2.1
untitled4.m 1
untitled4.m 2.2
Workspace
Name Value Size Class
Efficiency [2,4,6,8] 1×4 double
Modulat... 1×4 cell 1×4 cell
SNR [5,10,18,25] 1×4 double
untitled11 *
untitled12 *
untitled13 *
untitled14 *
untitled15 *
Efficiency = [2 4 6 8];
disp('Required SNR (dB) and Spectral Efficiency')
disp(table(Modulation',SNR',Efficiency', ...
'VariableNames',{'Modulation','SNR_dB','Efficiency'}))
figure
% -------- Graph 1 --------
subplot(2,1,2)
plot(SNR,Efficiency,'-o','LineWidth',2)
title('SNR vs Spectral Efficiency')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Spectral Efficiency (bits/s/Hz)')
grid on
% -------- Graph 2 --------
subplot(2,1,1)
plot(1:4,SNR,'-s','LineWidth',2)
xticks(1:4)
xticklabels(Modulation)
title('Required SNR for Different Modulation Schemes')
xlabel('Modulation Scheme')
ylabel('Required SNR (dB)')
grid on
Figure 1
Required SNR for Different Modulation Schemes
Required SNR (dB)
QPSK
16-QAM
64-QAM
256-QAM
Modulation Scheme
SNR vs Spectral Efficiency
Spectral Efficiency (bits/s/Hz)
Signal-to-Noise Ratio (dB)
Command Window
New to MATLAB? See resources for Getting Started.
{'64-QAM' } 18 6
{'256-QAM'} 25 8
>>
30°C
Partly cloudy
Search
ENG
IN
20:45
01-09-2026